

Supporting information of

A Machine learning-based hierarchical surrogate model of SWAT for computationally intense BMP optimization applications

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S1 preliminary experiments of Hierarchical Surrogate Model

Table ST1 preliminary experiments results

Model	Metric	RF	LSTM	ANN	SVM
Sub-watershed Surrogate model	Test Data R ²	0.9983	0.8754	0.9129	0.9842
	Test Data RMSE	0.0619	0.1319	0.0832	0.0698
	Training Time	5s	160s	80s	1540s
	Model File Size	120MB	60MB	5MB	50MB
In-stream Process Surrogate model	Test Data R ²	0.9457	0.9042	0.9943	-
	Test Data RMSE	0.1987	0.2680	0.1048	-
	Training Time	35s	1508s	2154s	-
	Model File Size	2.54GB	640MB	29MB	-

During the preliminary experimental phase, we extracted outflow simulation data from each sub-watershed and trained multiple machine learning methods. The results are summarized in Table ST1. In-stream model via SVM was not reported as they were not able to converge within 2 hours. Based on the above pre-experimental results, we found that the RF algorithm and the ANN algorithm have obvious advantages when they are respectively used to train the sub-watershed surrogate model and the river process surrogate model.

(1) Sub-watershed Surrogate Model Selection (RF):

Performance: RF achieved the highest R² (0.9983) and the lowest RMSE (0.0619) among all methods, demonstrating superior accuracy.

Efficiency: Training time for RF was 5 seconds, significantly faster than SVM (1540s) and LSTM (160s).

Practicality: Despite a larger file size (120MB), RF balanced speed, accuracy, and computational cost, making it ideal for sub-watershed modeling.

(2) In-stream Process Surrogate Model Selection (ANN):

Performance: ANN achieved the highest R² (0.9943) and the lowest RMSE (0.1048) for simulating reach outflow dynamics.

Compactness: ANN produced the smallest model file (29MB), enabling rapid deployment and integration.

Trade-off: Although ANN required longer training time (2154s), its accuracy and lightweight nature justified the choice for in-stream processes.

For the sub-watershed surrogate models, both RF and SVM showed comparable fitting performance. However, RF was ultimately selected due to its shorter training time (5s vs. 1540s for SVM) and higher simulation accuracy (R²=0.9983 vs. 0.9842 for

SVM). For the in-stream process surrogate model, ANN demonstrated exceptional performance in simulating cross-sectional flow dynamics ($R^2=0.9943$, $RMSE=0.1048$). Additionally, its compact model size (29MB) and fast inference speed made it the optimal choice despite longer training times.

S2 Data Required for Training the Hierarchical Surrogate Model

First, it is necessary to create a document that encompasses the information of each sub-watershed. This document serves the purpose of querying the fundamental information of sub-watersheds during the preparation of training data, and it should be named "Subwatershed_info.csv".

The following information should be included within this document (for specific details, please refer to the example provided in the attachment):

- SUB: the number of sub-watershed
- SUB_area_km2: Total area of the sub-watershed (km^2)
- mian_channel_km: Length of the main channel within the sub-watershed(km)
- Landuse: Proportion of the area occupied by various land use types (e.g., AGRL, RICE)
- Station: The number or name of the meteorological station utilized for the simulation of this sub-watershed (in this study, the number is employed as the name of the meteorological station)

Owing to reasons such as data confidentiality, the attachment only provides the format for generating the meteorological station data (refer to the attachment "59102.csv" for detailed information), and the specific data are not provided.

In accordance with the steps in the main text of the paper, following the construction of the SWAT model and the calibration of its parameters, the SWAT model should be run under different combinations of Best Management Practices (BMPs) parameters. Batch operations can be carried out with the assistance of software such as SWAT-CUP or external Python scripts. Corresponding output files will be generated during each SWAT simulation process. These files are as follows: a summary file (output.std), a Hydrologic Response Unit (HRU) output file (.sbs), a sub-watershed output file (.bsb), and a main stream or tributary output file (.rch).

Extract data from the SWAT model and organize it in the following format:

- 1.Naming: subDD_XX.csv (where "XX" should represent the number of the sub-watershed)
- 2.The data file should contain the following entries:
 - SUB: The number of sub-watershed

- YEAR: Simulation year
- MON: date
- PCP: Daily total precipitation within sub-watershed (mm)
- RH: Daily average relative humidity within sub-watershed (%)
- MAXTM: Daily maximum temperature within sub-watershed (°C)
- MINTM: Daily minimum temperature within sub-watershed (°C)
- WIND: Daily average wind speed within sub-watershed (m/s)
- BMPS: BMPs scenario setting number (which can be disregarded as it is not the data required for training)
- GW: Activation trigger of the grassed waterway (0 or 1)
- FSW: Filter strip width (m)
- FSC: Ratio of field area to filter strip area (ha/ ha)
- FRT_off: Fertilizer reduction ratio (%)
- WYLDmm: Surface runoff depths (mm)
- SYLDt_ha: Sediment yield (kg/ha)
- SubTN: Total Nitrogen loads (in kg/ha) (calculated by summing up the data of "ORGNkg_ha", "NSURQkg_ha", "LAT_Q_NO3kg_ha", and "GWNO3kg_ha" from the SWAT simulation outputs)
- FLOW_OUTcms: Average daily streamflow out of reach time step (m3/s)
- SED_OUTtons: Sediment transported with water out of reach time step (metric tons)
- TOT_Nkg: Total nitrogen transported with water out of reach (kg)

Among these, "PCP, RH, MAXTM, MINTM, WIND" are meteorological-related data, which can be obtained by querying the meteorological station corresponding to each sub-watershed in the SWAT input file. "GW, FSW, FSC, FRT_off" are BMPs parameters, which are set by the user. "WYLDmm, SYLDt_ha, SubTN" are simulation data at the sub-watershed scale, which can be found and calculated from the sub-watershed output file (.bsb) of the SWAT model. "FLOW_OUTcms, SED_OUTtons, TOT_Nkg" are simulation data of the main stream and tributaries of the river channel, which can be retrieved from the main stream or tributary output file (.rch) of the SWAT model.

Once the training data of all sub-watersheds have been organized, name them correctly and save them as ".csv" files for subsequent extraction during the training of the surrogate model. Please refer to the training data example "sub_DD_01.csv" in the supplementary materials for additional details.

S3 Training of the hierarchical Surrogate Model

The training and invocation codes of the model are provided in the attachment. The attached code encompasses hyperparameter settings for training scripts of various surrogate models, along with their corresponding training termination criteria. Comprehensive annotations have been incorporated throughout the scripts to explicitly document its specific functionalities, thereby enhancing code interpretability for other readers. Please execute them after ensuring that the data preparation work has been completed. The codes involve a series of operations, including the generation of training data and the verification of the model's performance. Some data examples used for training the surrogate model are provided, along with specific comments on the codes for reference.

Attachments:

The examples and codes used for constructing the Hierarchical Surrogate Model are available via GitHub (<https://bgithub.xyz/TOPGAN-Tifosii/Hierarchical-Surrogate-Model.git>).

1. Subwatershed_info.csv
2. Example of meteorological station data: 59102.csv
3. Example of hierarchical surrogate model training data: sub_DD_01.csv
4. Training script for the sub-watershed surrogate model: Subwatershed surrogate model.ipynb
5. Training script for the river channel process surrogate model: River channel process surrogate model.ipynb
6. Invocation script for the hierarchical surrogate model: Invocation code for the hierarchical surrogate model.ipynb
7. Example of the sub-watershed (number 01) surrogate model: SMFsub01.pkl
8. Example of the river channel process surrogate model: RCH_SM.h5